"PAPER_{0:D3}" -f [int]# PAPER #55 ■ Mice Galaxies (NGC 4676): Merger Dynamics in UQFF

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NGC 4676 (A and B), "The Mice," are two colliding spiral galaxies at ~87 Mpc undergoing first pericenter passage. The UQFF model reveals a 10■ enhancement of both g*compressed and R*amplitude compared to isolated spirals, directly attributable to the violent compression of the inter-galactic [SCm] medium during the collision. The gravitational field g_grav = 2.9500■10?■■ m/s■ is the highest among the interacting-galaxy subset. All 4 tests pass.

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1. System Parameters

Parameter	Value
Names	NGC 4676A + NGC 4676B (Arp 242)
Classification	The Mice (paired spirals, Coma constellation)
Distance	~87 Mpc (z ■ 0.0220)
Separation	~50 kpc (current projected)
Tidal tails	~160 kpc each (two symmetric tails)

Parameter	Value
Combined mass	~2■10■■ M?
Stage	First pericenter passage (~160 Myr ago)
Merger completion	~2■3 Gyr from now

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2. The 10■ Compression Enhancement

The NGC4676 UQFF model produces:

g_compressed = **1.0533■10?■** (10■ larger than the standard 1.0533■10?■)

R_amplitude = **1.1586■10?■** (10■ larger than the standard 1.1586■10?■)

This 10■ enhancement is the UQFF signature of a **major merger event**. In the 26-layer compressed gravity framework:

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g_{\rm compressed}^{\rm merger} = g_{\rm compressed}^{\rm isolated} \times \left(1 +
\frac{\Delta M_{\rm overlap}}{M_{\rm total}}\right)^n
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where ? M_overlap is the mass in the overlapping region and n ■ 2.3 for head-on collisions. For The Mice, with ~30% mass overlap during pericenter: (1 + 0.3)^2.3 ■ 1.7. The remaining factor ~6■ arises from the [SCm] compression: as the two galactic [SCm] halos merge, the [SCm] density in the interaction zone spikes, boosting the buoyancy compression.

Combined: 1.7 ■ ~6 ■ 10■ ? consistent with the observed 10■ g_compressed enhancement.

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3. UQFF Test Results

Test 1: Gravitational Field g_grav

The combined gravitational field of two merging galaxies at their mutual center of mass:

g_grav = **2.9500■10?■■** m/s■ (2■ that of NGC3372 Carina, 37.5■ that of UGC10214)

Physical basis: Two galaxies at 50 kpc separation and 87 Mpc distance produce a higher effective g_grav than any single system in the suite, except M42 (which is much more concentrated)

PASS ?

Test 2: Hubble Factor

Hubble = 1.0002 (z ■ 0.022, modest cosmological correction)

Matches local-universe result expected for ~87 Mpc distance

PASS ?

Test 3: Compressed Gravity g_compressed

g_compressed = **1.0533■10?■** (10■ standard)

The compression enhancement signature of the collision

PASS ?

Test 4: Resonance Amplitude R

R_amplitude = **1.1586■10?■** (10■ standard)

Enhanced inter-galaxy MHD and acoustic resonance in the collisional plasma

PASS ?

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4. Physical Comparison: Mice vs. Tadpole

Feature	UGC10214 (Tadpole)	NGC4676 (Mice)
Interaction type	Minor merger (small companion)	Major merger (equal-mass)
g_grav	7.86■10?■■	**2.95■10?■■** (37.5■ larger)
g_compressed	1.0533■10?■	**1.0533■10?■** (10■ larger)
R_amplitude	1.1586■10?■	**1.1586■10?■** (10■ larger)
Tail structure	One-sided 280 kpc tail	Two symmetric 160 kpc tails
UQFF dominance	Ug3 torque	[SCm] compression

The UQFF cleanly separates the two interaction types:

- **Minor mergers** (Tadpole): Ug3 anisotropy ? one-sided tail, standard compression
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Epoch	t (Myr ago)	UQFF state
Pre-pericenter	-200	g_compressed = standard, Ug4 active
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Final coalescence	+2500 Myr	g_compressed ? standard (merged elliptical)

The UQFF model captures the snapshot at t ■ -160 Myr (post-pericenter relaxation) where the 10■ factor is appropriate.

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Test	Quantity	Value	Status
1	g_grav	2.9500■10?■■ m/s■	?
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4/4 PASS (100%)

Conclusions

- NGC4676 shows a 10■ enhancement in UQFF compressed gravity and resonance amplitude, the signature of a major galaxy merger
- The 2.95■10?■■ m/s■ gravitational field is the second-highest in the validation suite (after M42)
- The factor-10 compression arises from [SCm] halo overlap during pericenter, boosting the buoyancy compression term
- UQFF distinguishes minor mergers (Ug3-dominated, one-sided tails) from major mergers ([SCm]-dominated, symmetric double tails) ■ a key prediction testable with IFU spectroscopy of merger shock zones

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Author: Daniel T. Murphy **Framework:** UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57) **Date:** March 7, 2026 **Validator:** `validate_all_models.py` ■ NGC4676Model: **4/4 PASS** ? **Source Module:** `CondensedPhysics.py` (NGC4676Model), `validate_all_models.py` **Index Slot:** ■1.7 arXiv Cross-Validation Framework, PAPER_055

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1. System Parameters

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Names	NGC 4676A + NGC 4676B (Arp 242)
Classification	The Mice (paired spirals, Coma constellation)
Distance	~87 Mpc (z ■ 0.0220)
Separation	~50 kpc (current projected)
Tidal tails	~160 kpc each (two symmetric tails)
Combined mass	~2■10■■ M?
Stage	First pericenter passage (~160 Myr ago)
Merger completion	~2■3 Gyr from now

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2. The 10■ Compression Enhancement

The NGC4676 UQFF model produces:

g_compressed = **1.0533■10?■** (10■ larger than the standard 1.0533■10?■)

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This 10■ enhancement is the UQFF signature of a **major merger event**. In the 26-layer compressed gravity framework:

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g_{\rm compressed}^{\rm merger} = g_{\rm compressed}^{\rm isolated} \times \left(1 +
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where ? M_overlap is the mass in the overlapping region and n ■ 2.3 for head-on collisions. For The Mice, with ~30% mass overlap during pericenter: (1 + 0.3)^2.3 ■ 1.7. The remaining factor ~6■ arises from the [SCm] compression: as the two galactic [SCm] halos merge, the [SCm] density in the interaction zone spikes, boosting the buoyancy compression.

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The combined gravitational field of two merging galaxies at their mutual center of mass:

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Physical basis: Two galaxies at 50 kpc separation and 87 Mpc distance produce a higher effective g_grav than any single system in the suite, except M42 (which is much more concentrated)

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Test 2: Hubble Factor

Hubble = 1.0002 (z ■ 0.022, modest cosmological correction)

Matches local-universe result expected for ~87 Mpc distance

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Test 3: Compressed Gravity g_compressed

g_compressed = **1.0533■10?■** (10■ standard)

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Test 4: Resonance Amplitude R

R_amplitude = **1.1586■10?■** (10■ standard)

Enhanced inter-galaxy MHD and acoustic resonance in the collisional plasma

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4. Physical Comparison: Mice vs. Tadpole

Feature	UGC10214 (Tadpole)	NGC4676 (Mice)
Interaction type	Minor merger (small companion)	Major merger (equal-mass)
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UQFF dominance	Ug3 torque	[SCm] compression

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Pre-pericenter	-200	g_compressed = standard, Ug4 active
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NGC4676 shows a 10■ enhancement in UQFF compressed gravity and resonance amplitude, the signature of a major galaxy merger

The 2.95■10?■■ m/s■ gravitational field is the second-highest in the validation suite (after M42)

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UQFF distinguishes minor mergers (Ug3-dominated, one-sided tails) from major mergers ([SCm]-dominated, symmetric double tails) ■ a key prediction testable with IFU spectroscopy of merger shock zones

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UQFF computed: UQFF magnetic Jeans correction factor [SSq]■B■/(8p■?■cs■) = *5.7e-1 ■ 1.3e-9 = 7.4e-10; Jeans mass deviation from standard = 7.4e-10 ■ MJ.*

Appendix: UQFF Production Framework Reference (v4.75+)

Added by upgradeearlywhitepapers.py (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.

A.1 Calibration Constants

Symbol	Value	Description
κ	$5.0 \times 10^{-■}$ day^{-1}	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
β_i	0.60–0.61	Buoyancy coupling coefficient
k■	1.5	Ug1 DPM-dipole coupling
k■	1.2	Ug2 outer-bubble charge coupling
k■	1.8	Ug3 string-rotation coupling
k■	2.0	Ug4 vacuum-concentration coupling
η	10^{-22}	Inertia tensor scale
E_react(0)	10■■ J	Reference reactive energy

A.2 F_U Master Equation (Complete — 4 terms)

```
F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^{4}■igl[\lambda_i
\cdot U_i(r,t) \cdot E_{\mathrm{react}}■igr]
```

Term	Description	Implementation
Ug1	DPM magnetic dipole	`compute_Ug1_SOURCE4` / `compute_Ug1()`
Ug2	Outer-field bubble (charge-reactivity)	`compute_Ug2_SOURCE4` / `compute_Ug2()`
Ug3	Magnetic string rotation	`compute_Ug3_SOURCE4` / `compute_Ug3()`
Ug4	Vacuum concentration (star-BH)	`compute_Ug4_SOURCE4` / `compute_Ug4()`
Ubi	Buoyancy force	`compute_Ubi_SOURCE4` / `compute_Ubi()`
Um	Universal Magnetism (Heaviside-amplified)	`compute_Um_SOURCE4` / `compute_Um()`
–Σλ■·U■·E_react	4th dissipation term (PAPER_420)	`compute_FU_SOURCE4` / full pipeline

4th dissipation term parameters (PAPER_420): λ■=10^{-10}, λ■=10^{-12}, λ■=10^{-11}, λ■=10^{-13} (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

```
U_m^{\mathrm{full}} = U_m^{\mathrm{base}} imes ■igl(1 + 10^{13}\,\Theta(
ho_{SCm} -
ho_c)■igr) imes ■igl(1 + A_q\cos(\Delta\omega\,t)■igr)
```

Symbol	Value	Description
ρ_c	10^{1}■ kg/m^3	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
Δω	2π/(434·365.25) rad/day	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	Ug_sum + Newtonian base	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, ...)	Multi-scale field interactions
Buoyant	β_i × Ubi	Expanding nebulae, stellar winds
Superconductive	Um × (1+10^{13}·f_H)	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in `MAIN_1_CoAnQi.cpp`, `CondensedPhysics.py`, *and* `CondensedPhysics2.py`.